FINAL EXAMINATION

Course: **Discrete Structure**	Class: **21TT1**
Time: **120 minutes**	Term: 1 – Academic year: 2022-2023
Lecturer(s): **Nguyễn Tấn Trung**	
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(Notes: Books and phones are NOT allowed)

<Followings are the questions and/or requirements>

Question 1: (2.5 pt) Draw the Hasse diagram (0.5 pt) for divisibility " | " on the set

$$(\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}, |)$$

a) (0.5 pt) Find the maximal and minimal elements.

b) (0.5 pt) Find the greatest and the least elements if it exists?

c) (0.5 pt) Find all upper bounds and the least upper bound (if it exists) of $\{2, 9\}$.

d) (0.5 pt) Find all lower bounds and the greatest lower bound (if it exists) of $\{60, 72\}$.

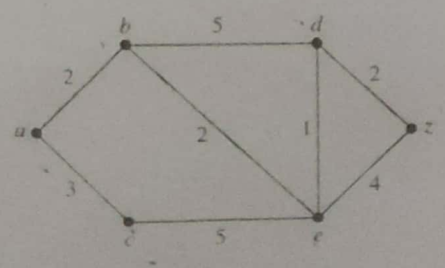
Question 2: (2 pt) Find a minimal expansion as a Boolean sum of Boolean products of the function in the variables of w, x, y and z

$$wxyz + wxy\bar{z} + wx\bar{y}z + w\bar{x}yz + w\bar{x}y\bar{z} + \bar{w}xyz + \bar{w}\bar{x}yz + \bar{w}\bar{x}y\bar{z} + \bar{w}\bar{x}\,\bar{y}z$$

a) (1 pt) Use a K-map.

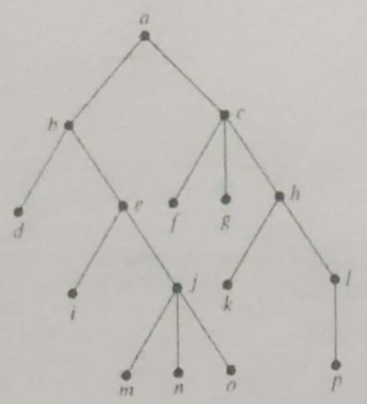
b) (1 pt) Use the Quine-McCluskey method.

Question 3: (1 pt) Use ***Dijkstra's algorithm*** to find the shortest path between "*a*" and "*z*". What is the complexity of ***Dijkstra's algorithm***? Explain your solution.



Question 4: (2 pt)

a) (1.5 pt) Determine the order in which a(n) preorder, inorder and postorder traversal visits the vertices of the given ordered rooted tree



b) (0.5 pt) What is the value of the postfix expressions

$$3\ 2 * 2 \uparrow 5\ 3 - 8\ 4 / * -$$

Question 5: (1 pt)

a) Find a recurrence relation for the number of **ternary strings** of length n that contain two consecutive 0s.

b) What are the initial conditions?

Question 6: (1 pt) Use generating functions to solve the recurrence relation $\boldsymbol{a_k = 5a_{k-1} - 6a_{k-2}}$ with the initial condition $\boldsymbol{a_0 = 6}$ and $\boldsymbol{a_1 = 30}$.

Question 7: (0.5 pt) Show that if G is a weighted graph and e is an edge whose weight is smaller than that of any other edge, then e must belong to every minimum weight spanning tree for G

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